System of first order linear ODEs for the Lorenz attractor.

$dx/dt = \sigma(y-x)$
$dy/dt = x(\varrho-z)-y$
$dz/dt = xy - \beta z$

What does it model?

The system was developed to be a simplified mathematical model for atmospheric convection. Meaning the original point of the model was to model the way that air moves when heated from below and cooled from above.

The system has three variables, x, y, and z. x is proportional to the rate of fluid flow, y is proportional to the temperature difference between the rising and falling air currents, and z is proportional to the distortion of the vertical temperature profile from the linear one. As the fluid is assumed to circulate in vertical and horizontal dimensions.

The system consists of three first order ODEs where each is a variable (x,y,z) differentiated with respect to t, representing time.

Meaning of the parameters?

The system has three parameters β, σ, and ϱ. β relates to the physical dimensions of the fluid layer. σ is the Prandtl number, and ϱ is the Rayleigh number.

The Prandtl number is the ratio of momentum diffusivity to thermal diffusivity. The Rayleigh number is the buoyancy driven flow or the free convection of a fluid.

The parameters determine whether the system is predictable or chaotic. A provided example for a chaotic lorenz system is where $\beta = 8/3$ $\sigma = 10$ $\varrho = 28$. The chaotic systems' behavior is hard to predict long term. Where the predictable systems' behavior are easy to describe.

For ϱ, the Rayleigh number, small values of ϱ usually will mean that the system is stable. When ϱ is larger ($\varrho > 24.74$) the two fixed points repulse the trajectory leading to the system being complex. The trajectories of the system (there are three) become uncorrelated after t = 24

Cite:

Wikipedia. *Lorenz system*. https://en.wikipedia.org/wiki/Lorenz_system